CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

Nov 2013

IT405 Advanced Computing

Date: 23.11.2013, Saturday **Time: 10:00 am To 01:00 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is cloud computing? List and Explain the roots of cloud computing? [05]

(b) Write the limitations of cloud computing application. [02]

Q - 2 (a) What is virtualization? Explain any three types of virtualization in brief. What is Full virtualization and Para virtualization? Write the difference between Full virtualization and Para virtualization. [04]

(b) Explain the cloud computing deployment model with example. List and explain the Characteristics of Cloud Computing. What are the pros and cons of cloud computing? [07]

(c) "Para-virtualization provides superior performance over full server virtualization." Justify the statement. What is hypervisor? [03]

OR

Q - 2 (a) What is Xen hypervisor? What are the purposes of Domain 0 and Domain U in Xen virtualization? [04]

(b) Explain the cloud computing delivery model with example. Explain the deployment of cloud computing in brief. [07]

(c) What is hypervisors? Write the differences between type 1 hypervisors and type 2 hypervisors? What is hypervisor? Explain desktop virtualization and application virtualization. [03]

Q - 3 (a) What is MPICH cluster? Explain the role of /etc/hosts file and shared directory in it. Write down the steps to Setup compute cluster using MPICH library in Ubuntu. [07]

(b) Google App Engine provides the facility of Platform as a Service (PaaS). Justify the statement. List the steps to deploy application on Google App Engine. What is Google App Engine? List and explain the functionality provided by Google App Engine? [07]

OR

Q - 3 (a) What is Map Reduce programming? Explain the purpose of scaling out and combiner function with example. What is Hadoop? Explain how Hadoop is used in distributed computing? [07]

(b) What is job tracker and task trackers in Hadoop Distributed File System (HDFS)? List the Areas where HDFS is not a good fit. Write the functionalities of name node and data node in Hadoop. [07]

SECTION – II

Q - 4 Define cluster computing and explain its components with an architecture diagram in detail. [07]

Q-5 (a) (i) There are five nodes in a cluster with response time of 10,20,30,40 and 50 nano seconds respectively. To expand the cluster, 5 more nodes with response time 20,40,60,80,150 nano seconds are added. After adding new nodes, will the performance of system improve or degrade as a whole? [04]

(ii) Differentiate between cluster, grid and cloud computing. [03]

(b) Explain the role of RMS (Resource Management and Scheduling) technique in Resource Scheduling in cluster computing. [07]

OR

(b) Explain HPVM (High Performance Virtual Machine) and CLUMPS (Clusters of SMP) with their advantages. [07]

Q-6 (a) (i) Why Grid is consider as next generation internet? How Grid provides restricted access to resources? [04]

(ii)List major grid computing methods and discuss any one. [03]

(b) Explain the role of virtual organization in grid computing. [07]

OR

(b) Discuss in detail implementing production grids. [07]